Supplementary Material

OUYANG Tiancheng
Independent Researcher
nighty@nightytech.com
ORCID: 0009-0002-0335-7147

November 11, 2026

Microphone Selection and Recognizer Cascading for Embedded Meeting Transcription: An RK3588 Study

This supplement describes the experiments, exact policy settings, paired four-policy attribution, component costs, and reproducibility boundaries. The system-level synthesis is retrospective; original frozen comparisons, late secondary analyses, post-unblinding supplementation, and subsequent frozen-policy evaluation retain their separate statistical identities. The audit/minimal follow-up in S18 comprises 2,400 actual requests and 1,200 exact product pairs on the exposed timing subset, with no additional independent quality sources.

The primary extraction sources in the author's archive are `DATA_CODE_AND_EVIDENCE.md`, `SUPPLEMENTARY_CONTROL_PARAMETERS.md`, `configs/POLICY_PARAMETERS.json`, `RESEARCH_SUMMARY_ZH.md`, and `PAPER_DRAFT.md` in `system_empirical_paper_20261016`.

S1. Evidence locations and reading conventions

The public companion repository in S10 contains numerical scores and timings, analysis code, configurations, and asset descriptors. It supports numerical recomputation through the command in S9. Transcripts, raw native tensors, SDK binaries, execution logs, and the recovered conversion log are retained only in the author's archive, including research-device storage; they are excluded from the public export.

The author's complete study archive is separate from the public export and is not a supplied manuscript attachment. The older experiment-directory names below identify its historical study root, hereafter **ROOT**. Other local analysis and execution paths identify records in the author's revision archive unless explicitly marked as public.

The following abbreviations identify exact directories, not public repositories:

Abbreviation	Directory relative to ROOT
M	`system_empirical_paper_20261016`
A	`spatial_independent_test_20261015`
F	`fixed_channel_extension_20261016`
C	`calibration_independent_test_20261016`
S	`spatial_distinct_probe_confirmation_20261016`
P	`spatial_distinct_probe_confirmation_20261016/completed_capture_v2/package`
D	`probe_choice_mechanism_20261016`

An entry such as `P/analysis_01/QUALITY_ROWS.jsonl` expands using this table. Recorded device paths are provenance identifiers; their appearance in metadata does not establish that the binary payload is included in the public export.

S2. Cohorts, sampling units, and execution counts

Table S1. Data and execution denominators. A source is one selected speech interval. A policy cell is a policy execution or explicitly labeled projection for that source. Native calls can exceed policy cells because a source may be split into several parts.

Phase	Quality and selection data	Timing or additional execution	Sampling and inferential boundary
AliMeeting development	24 intervals from four meetings; channel 4 selected as the original static comparator	Development comparisons are separate from Test	Small development selection; ch4 is not asserted to be the globally best microphone
AliMeeting original	600 Test intervals from 18 recordings; 13,990 normalized reference characters; 2,898.6 s; nine policies	Preselected 36-source panel from the same 18 recordings, two measured rounds; original full-cohort one-pass timings retained separately	Paired whole-recording inference; repetitions and policies do not add independent quality samples
AliMeeting EV secondary	The same 600 sources; EV and contemporary RMS control	Same 36-source panel, two new timing rounds	Late secondary protocol frozen before its quality unblinding; use contemporary EV/RMS costs
AliMeeting all-fixed supplement	The same 600 sources; six formerly missing fixed channels plus fresh ch0/ch4/RMS anchors	5,400 actual policy cells, including 1,800 anchor cells; fresh-phase cost records kept separate	Post-unblinding supplement; all eight static channels become available, with no new independent quality N
AISHELL-4 startup calibration, C	20 recordings, 20 separate startup prefixes C1, and 800 later Test intervals (40 per recording); 20,394 Test reference characters; five policies	Preselected 40 later sources, two timing rounds, plus repeated startup setup; final report records 4,880 own-policy cells and 5,669 native SV calls across the three runs	Recording is the resampling unit; C1 is excluded from Test CER; Qwen was initialized but never called for recognition
AISHELL-4 small-room evaluation, S	600 intervals, 20 per recording, from 30 `train_S` recording files; 13,498 reference characters; nine dates and one filename-R label, R001	Quality bank: three actual query types per source, 1,800 bank-query cells. Cost: 120 preselected sources, four per recording, ten policies, two rounds, 2,400 actual policy cells	Ten-policy quality is response-bank projection; recording resampling is conditional on observed date×R strata; 600 sources are not 600 iid observations
S ch0-to-RMS comparison	Reuses the same 600 sources and existing real ch0-SV/RMS-SV responses	No new recognition or ch0-policy service measurement	Posthoc descriptive comparison of an exposed cohort; no additional independent confirmation
Old probe-choice mechanism screen	Reuses the exposed AliMeeting 600-source data; ten previously enumerated rules and existing confidence curves	Existing response/count calculations; no new service measurement	Development/mechanism evidence only; predicted query counts are not elapsed cost

AliMeeting selection preceded recognition. Approximately balanced recording quotas and deterministic speaker-stratified selection retained eligible complete, non-overlapping 2–15 s intervals. Of 20 Test meetings, 18 supplied eligible sources; the two exclusions followed the eligibility rule rather than observed recognition quality. The six filename room-prefix groups and 48 recording-local speaker labels do not establish six independent physical rooms or 48 distinct people.

For C, C1 is the earliest eligible interval in each recording. The 40 later intervals per recording were selected from chronological quartiles before recognition. E1 and L1 commit their startup choices before later Test recognition; only L1 consumes a C1 reference transcript. Later Test references are never runtime inputs.

For S, exposure exclusions were applied before new recognition, including historically recognized `train_L` sources. Sampling retained recording-local speech-tier coverage and seeded selection. The archived protocol specifies recording-selection seed 2026101627 when subsampling eligible recordings and quality-source seed 2026101621 with per-recording streams; the separate pre-recognition cost selection is bound by the saved manifests and selection code. Exact streams, ordering, frame coordinates, and exclusions are authoritative in `P/formal_preparation_01/select_formal_sources.py`, its preparation records, and `P/input_materialization_01/SOURCE_MANIFEST.json`. The supplied `train_S` name describes the corpus partition; these S outputs were not used to fit or choose the frozen routing policies. Historical study-exposure checks do not determine exposure during foundation-model pretraining.

Thirty recording files are not thirty verified independent rooms or participant groups. Physical-room identity, cross-file participant dependence, and model-training exposure remain unresolved. Across studies, differences in recording conditions, content, segmentation, and possible model exposure prevent a cross-corpus causal attribution to one acoustic factor.

S3. Device, recognizers, and software

Throughout the cost ledgers, "Teacher" denotes Qwen3-ASR-0.6B and "outer" denotes its enclosing call timer, not teacher–student training. Nested SDK timers are not additional additive components.

Table S2. Recorded deployment configuration.

Component	Recorded setting and limit
Device	Orange Pi 5 Max, RK3588, nominal 16 GB RAM; approximately 15.59 GiB OS-visible RAM; eight logical CPUs
CPU observations	Little-core snapshots reached 1.8 GHz and big-core snapshots 2.352 GHz. These are observed snapshots, not locked clocks or complete historical scaling caps
Qwen decoder CPU configuration	Logical CPUs 4–7, mask `0xF0`; `enabled_cpus=4` must not be interpreted as little cores 0–3
Residency	Both recognizers remained loaded; service does not characterize optimized single-model memory use
SenseVoice	SenseVoice-Small, frozen RKNN preprocessing/tokenizer/greedy CTC decoder; four prefix output frames omitted before extracting valid speech-frame logits
SenseVoice splitting	16 kHz input; sources longer than 152,000 samples use the frozen low-energy splitting procedure; concatenate part transcripts. Waveform-dependent cut positions may differ between input policies
Qwen	Qwen3-ASR-0.6B; available fixed-30 s encoder and RKLLM decoder; deterministic top-1 decoding; maximum generation 256 tokens; repetition limit 257
Qwen context	Clear decoder context on every call; no preceding-source history, reference transcript, or textual hints
Native runtimes	Recorded identities include RKNN 2.3.2 and RKLLM 1.2.3; exact stage-specific graphs, binaries, libraries, and configurations are hash-bound
Precision	Components have different precisions; the complete pipeline is not described as uniformly FP16

The Qwen repetition setting corrects an earlier implementation stop rule and is shared across current conditions. It is not a new recognition method. Measured Qwen/SV service ratios concern these two exports, including the fixed encoder extent; they do not identify an architecture-inherent or NPU-inherent speed ratio.

The AliMeeting reproducibility record distinguishes the canonical local analysis environment (Python 3.14.7, NumPy 2.5.3, Unicode 16.0.0; main figure environment Matplotlib 3.11.2), device environment (Python 3.12.13, NumPy 2.4.4, SciPy 1.17.1), and EV-port equivalence environment (Python 3.12.14, torch/torchaudio 2.8.0). These phase-specific records do not imply that all phases used one environment. Corresponding runtime/configuration records govern C and S. A direct-dependency list is not a complete wheel/OS lock; pip packages alone do not reproduce the native SDK environment.

Exact export and runtime identities can be located through `A/prepared_01/MODEL_ASSET_BASELINE.json`, each original run's `native_v1/RUNTIME_BINDINGS.json`, `C/prepared_attempt_01/MODEL_ASSET_BASELINE.json`, and `P/formal_01/{bank_01,timing_01,timing_02}/native_v1/RUNTIME_BINDINGS.json`.

Recovered provenance. Files recovered on October 22, 2026 identify the upstream model as Qwen/Qwen3-ASR-0.6B. All six principal deployment assets match the recorded device hashes; all 311 repackaged decoder tensors match their upstream counterparts. The public decoder script and the successful log retained in the author's archive specify RKLLM Toolkit 1.2.3, no quantization, optimization level 1, RK3588, three NPU cores, and maximum context 4096. Encoder ONNX metadata records PyTorch 2.6.0, opset 17, and inputs $[1, 3000, 128]$ and $[1, 390, 896]$; historical model headers identify RKNN Toolkit2 2.3.2. The upstream revision and complete encoder export/build records remain unavailable. Precision filenames do not establish all internal operator precisions.

The author confirms original SenseVoice Small. Its exact checkpoint revision and historical conversion script, quantization settings, and calibration history were not recovered. Historical headers identify conversion Toolkit 2.3.3a25/compiler 2.3.3b3, distinct from runtime 2.3.2. The repository records hashes, recovered scripts, graph metadata, licenses, and remaining gaps. Identical re-export from upstream weights has not been demonstrated.

S4. Frozen input and startup processing

Table S3. Input recipes and exact implementation entries.

Recipe	Frozen behavior	Implementation entry
Static channel	Return the selected original channel	`A/prepared_01/frontends.py`; `F/prepared_01` for added channels
RMS	Choose the smallest-index maximizer of mean squared channel samples; return that raw waveform without amplitude normalization	`A/prepared_01/frontends.py` and each phase's frozen copy
Unaligned mean	Average all eight channels	`A/prepared_01/frontends.py`
GCC-PHAT	Integer delays relative to channel 0, search ±32 samples, zero-padded alignment and summation	`A/prepared_01/frontends.py`
Blind MVDR	512-sample STFT, hop 128; mixture-covariance steering; lowest-energy 20% of frames for noise covariance; trace-scaled diagonal loading; numerical guards fall back to ch0	`A/prepared_01/frontends.py`; exact loading and guard constants remain in the frozen code
EV	Existing envelope-variance recipe adapted to top-one raw-channel selection; 400-point STFT, 40 mel bands, archived compression/window rules	`A/ev_prepared_00/ev_frontend.py` and `EV_PROTOCOL_v1.md`; C frozen copy and `EV_PORT_EQUIVALENCE_v1.json`
E1 startup energy	Select highest-energy C1 channel once and reuse it on later Test sources	`C/prepared_attempt_01/calibration_rules.py` and `CALIBRATION_LABELS.json`
L1 labelled startup	Recognize all eight C1 channels; select lowest reference-relative edit count, then power, then channel index for ties; reuse on later Test	Same C frozen entries

GCC and MVDR are the specified frozen implementations; their results do not characterize an optimized beamforming system or a jointly trained recognizer. EV is not a reproduction of a complete GSS pipeline. Power need not rank target-speech quality correctly. Exact constants not enumerated in the synthesis are obtained from the identified frozen code, not substituted from a modern implementation or guessed defaults.

Unnormalized channel RMS depends on recording-chain gain and any automatic gain control (AGC), so its observed performance is conditional on the recorded channels. Startup channel selection does not calibrate electrical gain or establish robustness across gain/AGC settings.

S5. Ten frozen S policies

The first SV recognition and Qwen use the RMS waveform. An optional SV probe uses channel 0. When invoked, Qwen's complete output replaces the first hypothesis; no local hypothesis fusion or ROVER is used.

Table S4. Routing behavior.

Policy	Decision rule
First-only	Execute RMS-SV and keep its successful output, including successful empty output
Full-Qwen	Execute RMS-Qwen without preliminary SV or unnecessary SV confidence work
Mean95	Execute Qwen when mean entry confidence is below 0.95
Distinct probe	Below 0.90 execute Qwen; at or above 0.95 retain first output. In [0.90, 0.95), identical RMS/ch0 mono waveforms trigger Qwen; otherwise pay a ch0-SV probe and execute Qwen only on normalized-text disagreement
Always-probe	Compare first/probe normalized texts throughout; skip redundant same-wave recognition and invoke Qwen on disagreement
Legacy conditional	Same middle-band distinct-input probe rule, but same-wave identity is treated as agreement and retains the first output
Cheap same-view	In the middle band retain first output for identical inputs; otherwise invoke Qwen directly without paying for a probe; retain the low/high gates
Confidence mixture low	Use one component assigned before recognition, with parameters in Table S5
Confidence mixture matched	Same assignment mechanism, with its own frozen probabilities in Table S5
Gain RF	Execute Qwen when predicted signed edit gain exceeds 0.05 times the old-training mean teacher ASR cost

Successful recognition with no greedy entries escalates before numeric gating in dynamic arms. Missing or invalid arrays fail rather than silently becoming high confidence. Full-Qwen executes its named recognizer directly. The exact empty-output and identity branches are preserved in `P/formal_01/bank_01/formal_policy_core.py`, with the own-execution implementation in `formal_execution_core.py`.

For confidence, collapse consecutive greedy token IDs before removing blank ID zero, separately for each valid SV part. Use float64 softmax. Mean confidence pools nonblank greedy run-entry maximum posteriors across all parts. Shannon confidence is one minus summed normalized entry entropies divided by the total retained entry count; normalize each entropy by log vocabulary size. Neither score averages all valid frames or takes an unweighted mean of part means. The exact implementation is `P/formal_01/bank_01/confidence_metrics_v1.py`.

S6. Parameter provenance and assignment

All mixture values below are copied without rounding from `M/configs/POLICY_PARAMETERS.json`, whose locally checked SHA-256 is:

`7e72761b1d7620d04e6cc7c57f2adb01eec724ac521fc81331cdd70619329449`

The same asset is bound in `P/formal_01/STUDY_PIPELINE_CONFIG.json`. The synthesis did not change thresholds, weights, or methods.

Table S5. Frozen mixture components. Invoke Qwen when the assigned confidence is strictly less than its threshold. The parameters below reproduce the frozen control-parameter file; table numbering is local to this supplement.

Component, in cumulative assignment order	Threshold	Low probability	Matched probability
Normalized Shannon entry confidence	0.9753030446720764	0.863843176886491	0.472259298588883
Mean entry maximum posterior	0.962405133427767	0.004549891502801637	0.3897045689554038
Mean entry maximum posterior	0.9722072538148704	0.13160693161070738	0.13803613245571325

Assignment uses Python `random.Random(2026101619).random()`: in frozen quality-source order, draw one U per source, shared by the two mixtures, and apply each mixture's cumulative probabilities in table order. Zero-weight components are never selected; an exact cumulative boundary selects the next positive-weight component. Persist assignments before first new S recognition and reuse them in both timing rounds. Compute only the assigned component's recipe. Equal confidence values are never split by source ID. This is source-independent random assignment before observing outputs, not post-recognition best-of-three selection or a conjunction of three gates. The exact saved assignments are `P/formal_01/MIXTURE_ASSIGNMENTS.json` and the stage copies.

Mixture thresholds and working points were selected using complete old AliMeeting confidence-threshold groups, pooled/macro quality constraints, and component costs. Both working points are retained. No S-based seed selection or threshold fitting is claimed.

RF predicts signed edit gain (first errors minus Qwen errors) from eight features: mean, minimum, and tenth-percentile entry posterior; normalized output length; log duration; channel-RMS max/min; channel-0/max ratio; and whether RMS selected channel 0. Six capacities × seven cost multipliers were compared using old AliMeeting filename-room-prefix cross-fitting. Selection minimized old-development component cost subject to the revised development pooled/macro quality constraints. This tuning is not an independent out-of-fold test.

The selected RF uses 200 trees, depth 3, minimum leaf size 20, all features, seed 2026101624, and single-thread evaluation. Final fitting uses old data only. The threshold uses lambda 0.05 edits/s and old-training mean teacher ASR time 3.145955260156358 s, not held-out S time. The exact RF asset is `P/formal_preparation_01/policy_inputs_v1/CHEAP_RF.json`, also copied into each formal stage; SHA-256:

`2f64155d9c9f9aaf4f37a710b551d456e4d4c48292e2aebafd30f390e37e7fc2`

Feature transformations and exported-tree evaluation are specified in `P/formal_01/bank_01/rf_inference_v1.py` and `formal_execution_core.py`. The captured formal-preparation directory retains the selection record. No S labels were used to fit this model or choose its threshold.

The two mixture rows and their recording-bootstrap intervals condition on the one persisted component allocation. Repeating the same assignment controls paired execution but does not cover variability from drawing a new allocation. Small point-ordering differences between a mixture and another policy are not allocation-invariant.

S7. Quality and uncertainty

Primary normalization applies Unicode NFKC, casefolding, and retention of Unicode letters/numbers. No number-word rewriting, reference repair, or inverse text normalization is introduced. Use the archived normalizer and Unicode environment; changing normalization can change denominators and comparisons.

For recording r, let E_r be total Levenshtein edits and N_r the normalized reference-character count. Pooled CER (%) is $100 \sum_r E_r / \sum_r N_r$; recording-macro CER (%) is $100R^{-1} \sum_r E_r/N_r$. Source scores, recording aggregates, paired differences, and normalization-specific outputs remain separately inspectable. Percentage-point differences are distinct from relative-percent changes.

Table S6. Resampling and interval families.

Analysis	Unit and draws	Reported coverage and interpretation
Original AliMeeting primary contrasts	10,000 paired whole-recording draws	Marginal 97.5% intervals for two primary comparisons within an estimator. Pooled and macro families are not jointly protected
Other AliMeeting secondary/interaction results, unless explicitly labeled otherwise	Paired recording analysis in archived code	Marginal descriptive 95%, outside the original primary multiplicity protection
All-fixed supplement	Same 18-recording cohort, paired contrasts	Eight marginal 99.375% intervals per estimator/normalization; post-unblinding family
C primary calibration contrasts	50,000 paired recording draws	Marginal 98.75% intervals for four primary calibration comparisons; separate from descriptive timing inference
S quality and service	10,000 paired recording draws within fixed date×R strata, original stratum weights preserved; seed 2026101622	Marginal descriptive 95%; conditional on the observed strata, not nine-comparison familywise protection. One singleton stratum contributes no estimated within-stratum resampling variance
S ch0-to-RMS supplement	Same frozen S date×R engine and normalization	Posthoc descriptive 95%; no new independent sample or confirmation

Use each phase's saved analysis configuration and code for seeds and estimator details not enumerated here: `A/analyze_formal.py`, `A/analyze_timing.py`, `F/analyze_fixed.py`, `C/ANALYSIS_FREEZE_v1.json`, `C/analyze_calibration.py`, and `P/formal_preparation_01/formal_statistics.py`. These file entries are the specification; unspecified seed values are not invented in this supplement.

For service, average repeats within source before recording-level aggregation. S then averages sources within recording and recordings for the reported macro service. Preserve the same paired recordings and observed stratum weights in uncertainty calculations. Repetitions improve measurement coverage but do not create independent recordings. Leave-one-date/condition-out summaries are descriptive sensitivities of the same cohort.

An interval crossing zero does not establish equality, equivalence, or noninferiority. Separate quality and cost intervals do not constitute a joint dominance region. There is no cross-cohort pooled confirmatory test or universal ordering of input selection and recognizer upgrade.

S8. Warm-service timing and response-bank semantics

Service begins when the multichannel input buffer is available. It includes the required frontend, the policy's actual recognition calls, required confidence/features/RF evaluation and routing, and final text processing. S also accounts for necessary output-copy work. Initialization, shared source I/O, source-file verification, environment gates and waits, and disk archiving remain separate. In S, the outer policy clock encloses Qwen call interception, in-call input/integrity hashing, and evidence deep copies inside the helper. These operations are not all separately timed. They remain in the reported instrumented service; no guessed overhead is subtracted, and the measured savings are not presented as isolated deployment-only computation. This is buffered warm processing, not capture-to-endpoint delay, word-commit latency, continuous streaming performance, energy, or optimized single-model deployment cost.

Every actual policy cell owns its inference calls. Content-addressed storage deduplication occurs after execution and cannot replace a required call. A shared nonblocking lock and identity-checked guards prevent competing measurements, verify inactive recording, and restore paused processes. Balanced orders and two measured rounds reduce order confounding. S uses separate process initialization per timing round, reversed source/policy order in the second round, and recorded 75 °C/load1≤4 gates with waiting outside service. Governors were not changed. Historical affinity, scaling caps, and thermal behavior were not completely recorded; current snapshots must not be used to backfill old settings.

For S quality, each source supplies real RMS-SV, ch0-SV, and RMS-Qwen responses. Frozen policies select from this common bank and are scored offline. The resulting ten-policy table is **bank-projected quality**, not 6,000 separately executed routes, and quality rows have no assigned service. Predicted Qwen/probe counts over 600 sources describe routing selections, not measured elapsed service or actual SDK-call totals.

On the preselected cost subset, every policy executes its own causal routing and recognition twice without bank lookup. Text, action, empty-output branch, mixture assignment, and native/SDK counts are compared against bank predictions. All 2,400 checked cost cells matched on those fields. This bounds observed drift on that subset; it does not establish universal equivalence between bank projection and live execution.

For example, distinct probing and cheap same-view each made 108 Qwen SDK calls across the two actual rounds, but required 306 versus 254 native SV calls. Mean95 made 122 Qwen and 254 native SV calls. These are subset execution counts; the corresponding full-quality predicted counts in the main table use a different denominator. Multipart native calls must also be distinguished from source-level routing counts.

Do not merge original AliMeeting repeated times, contemporary EV/RMS repeated times, fresh all-fixed phase times, C times, or S times. Original one-pass times remain historical measurements rather than replacements for repeated-panel estimates. C's L1 startup components averaged approximately 2.635 s; human transcription latency and waiting for C1 were unknown. Those omitted quantities prevent a complete setup-amortization claim. Relative to RMS, E1 and L1 service-difference intervals cross zero. Relative to EV, they save 59.386 ms [45.085, 74.146] and 58.845 ms [46.814, 71.063] per later interval (marginal descriptive 95%); unresolved quality differences do not establish quality equivalence. No ch0 warm-policy service was measured or reconstructed for the posthoc S input comparison.

S9. Exact evidence and reproduction entry points

Table S7. Author-archive entry map. The entries below identify retained records and source code, not the public repository's file listing.

Scope	Exact entries, relative to ROOT through S1
AliMeeting sample reconstruction	`A/prepared_01/evidence/source_manifest.json`, `A/prepared_01/manifest.json`, `A/prepared_01/METHOD_FREEZE.json`; census/exposure/selection records indexed by `A/paper_materials_v2/DATA_AND_EVIDENCE_INDEX.md`
AliMeeting quality and original repeats	`A/analysis_v1/per_example.json`, `A/analysis_v1/per_example.csv`, `A/analysis_v1/VALIDATION.json`, `A/timing_analysis_v1`; original records in `A/attempt_01`, `A/timing_repeat_01`, `A/timing_repeat_02`
AliMeeting commands and environment	`A/paper_materials_v2/ORIGINAL_COMMANDS.md`, `A/paper_materials_v2/REPRODUCIBILITY.md`, `A/paper_materials_v2/METHODS.md`, `A/paper_materials_v2/OBSERVED_RUNTIME_v1.json`
EV	`A/ev_prepared_00/EV_PROTOCOL_v1.md`, `A/ev_prepared_00/ev_frontend.py`, `A/ev_secondary_00`, `A/ev_secondary_01`, `A/ev_secondary_02`, `A/ev_analysis_v1`; controller `A/continue_ev.py` and analyzer `A/analyze_ev.py`
All-fixed	`F/PROTOCOL.md`, `F/prepared_01`, `F/attempt_01`, `F/analyze_fixed.py`; `F/analysis_v1/per_source.csv`, `per_policy.csv`, `contrasts.csv`, `anchor_comparison.csv`, `fresh_phase_costs.csv`, `VALIDATION.json`, `SHA256SUMS.json`
C sample/configuration and actual runs	`C/PROTOCOL.md`, `C/PRE_RECOGNITION_FREEZE_RECEIPT_v1.json`, `C/ANALYSIS_FREEZE_v1.json`, `C/TIMING_PANEL_v1.json`; `C/prepared_attempt_01/{manifest.json,SOURCE_CERTIFICATE.json,CALIBRATION_LABELS.json,METHOD_FREEZE.json}`; collected records in `C/analysis_inputs_v1`

Scope	Exact entries, relative to ROOT through S1
C analysis and setup	C/analyze_calibration.py, C/analysis_v1/{ANALYSIS.json,per_source.csv,per_recording.csv,contrasts.csv,cost_per_cell.csv,timing_averaged_sources.csv,timing_averaged_setup.csv,calibration_costs.csv,main_repeat_drift.csv,VALIDATION.json,OUTPUT_SHA256.json}; C/calibration_diagnostics_v1/REPORT.md; C/FINAL_REPORT.md
S source identity and exposure	P/input_materialization_01/{SOURCE_MANIFEST.json,PROVENANCE.json,EXPOSURE_VERIFICATION.json,PLAN.json,COMPLETE.json}; P/formal_preparation_01/{select_formal_sources.py,HISTORICAL_EXPOSURE_v1.json,HISTORICAL_EXPOSURE_STRATA_v1.json,BEFORE_FIRST_NEW_ASR.json}
S freeze and policy execution	P/formal_01/{PACKAGE_FREEZE.json,PACKAGE_PLAN.json,STUDY_PIPELINE_CONFIG.json,MIXTURE_ASSIGNMENTS.json,PIPELINE_COMPLETE.json}; each of bank_01, timing_01, timing_02 contains SOURCE_MANIFEST.json, SCHEDULE.json, RUNTIME_CONFIG.json, run_native.py, formal_execution_core.py, formal_policy_core.py, confidence_metrics_v1.py, rf_inference_v1.py, POLICY_PARAMETERS.json, CHEAP_RF.json, STAGE_COMPLETE.json
S scoring, statistics, and drift	P/formal_preparation_01/{quality_projection_v1.py,analyze_formal_quality.py,formal_statistics.py,run_formal_analysis.py}; P/analysis_01/{QUALITY_ROWS.jsonl,QUALITY_STATISTICS.json,COST_ROWS.jsonl,COST_STATISTICS.json,ACTUAL_CALL_COUNTS.json,DRIFT.jsonl,DRIFT_REPORT.json,INPUT_BINDINGS.json,COMPLETE.json}
S posthoc input comparison	S/s_ch0_rms_descriptive_v1/{PLAN.md,POSTHOC_MANIFEST.json,analyze_ch0_rms.py,PER_SOURCE.jsonl,RESULTS.json,REPORT_BINDINGS.json,COMPLETE.json}
Old probe-choice mechanism	D/{PLAN.md,probe_choice_screen.py,PER_SOURCE.jsonl,RESULTS.json,REPORT.md}

Braces in this table enumerate separate files, not a literal filename. Stage-specific source copies and freeze records take precedence over generic filename similarity elsewhere in the workspace. Original scripts may contain absolute paths and exclusive-output constraints; a pathname alone does not make them relocation-safe.

For public numerical reproduction, install the dependencies listed in the repository's `requirements.txt` and run the following from its top-level directory:

```
python3 -B reproducibility/reproduce.py --bootstrap --output reproduced
```

The command checks the public integrity manifest and recomputes A/F/C/EV pooled and recording-macro CER summaries, S quality/cost estimates and bootstrap intervals, and the matched service summaries described in S18. It uses released edit counts and timings; transcripts are absent, so it cannot independently normalize or rescore text or re-establish text/action parity. It does not rerun the A/F/C/EV bootstrap, native tensor audits, inference, or the complete historical manuscript pipeline. The repository's reproducibility README specifies coverage and outputs; `reproduced/VERIFICATION.json` records the run status. The historical text-rescoring and S14 reconstruction scripts belong to the author's archive, not this public command.

Historical AliMeeting-only receipts in the author's `A/portable_release_v1` and `A/PORTABLE_RELEASE_README.md` cover a 13,200-row metadata/text check. That check is separate from public numerical reproduction and does not establish upstream model reconstruction or a C/S hardware rerun.

Original AliMeeting execution and analysis commands are retained in `A/paper_materials_v2/ORIGINAL_COMMANDS.md`. Hardware reproduction requires the separate payloads and native runtimes, adapted paths, and new output directories; the public numerical command performs no hardware rerun.

S10. Availability, licenses, and provenance limits

The evidence column below describes the author's retained study records. Public availability is limited to the numerical export and code specified in the companion repository; the table does not imply release of the complete archive.

Table S8. Distinct availability and reproducibility scopes.

Scope	Existing evidence	Boundary
Metadata, source code, configurations, scores, and traces	Frozen manifests, SHA bindings, source scores, response/routing records, audit receipts; AliMeeting portable bundle	Public code and score/cost reconstruction records at https://github.com/nighty35628/rk3588-meeting-asr; repository documentation specifies release contents and exclusions
Dataset reconstruction	Official OpenSLR 119 (AliMeeting) and 111 (AISHELL-4), with exact source IDs/partition coordinates and hashes in phase manifests	Obtain original distributions and reconstruct from manifests subject to corpus terms; this supplement neither downloads nor republishes audio
Corpus license	Corpus licenses and attribution are recorded in the official OpenSLR source checks supplied with the archive	Audio redistribution and derivative obligations must follow the applicable corpus terms; code/model licensing is separate
Raw audio and execution payloads	Archived indices for recordings, parts, input float matrices, native tensors, and returned strings; earlier physical replay/audit receipts	Large WAV/FLAC/raw payloads remain on the device and are not fully mirrored on the laptop; a visible descriptor path is not possession of its payload
Deployed model/runtime assets	Existing model/library fingerprints and configuration records	Artifact-level identity is documented; access rights and redistribution are not implied by a hash
Upstream reconstruction	Qwen checkpoint hash, matched deployment assets, recovered decoder script/log, and encoder graph metadata; SenseVoice author confirmation and historical runtime records	Exact revisions and complete encoder/SenseVoice conversion histories remain unavailable; complete re-export is not demonstrated
Historical study exposure	Selection/exclusion histories and S exposure verification	Prior study recognition is distinguished from catalogue access; pretraining exposure, cross-recording person identity, and physical-room independence remain unknown

The archived AliMeeting Test archive has 9,555,497,666 bytes and local SHA-256 `42407625b774a44fe34f2f25ca99116f090bec901f5ab81196b89f683a8b7bfa`. This is a local artifact fingerprint, not an official publisher attestation. AISHELL-4 container/member identities and exact selected intervals are specified by the phase source manifests; no replacement download URL or guessed checksum is supplied.

The AliMeeting reproducibility record reports 92,314,323,600 bytes of raw/processed payloads retained on the board across its six run directories. The synthesis separately reports 2,197 S metadata files individually SHA-checked locally. These historical checks describe the author's archive; they are not a public payload inventory or a new physical audit.

Publication availability: Numerical scores and timings, analysis code, configurations, and recomputation instructions are deposited at https://github.com/nighty35628/rk3588-meeting-asr. Its README specifies release contents and licensing; the author-only materials listed in S1 are excluded. Original audio is available through the official corpus distributions. Complete audio-to-results and upstream model-export reproduction remain incomplete.

S11. Counterexamples and interpretation limits

The all-fixed supplement matched fresh ch0/ch4/RMS anchor texts and scores but added no independent sources. RMS has the lowest point CER; its adjusted comparison with ch7 crosses zero. Test-best ch6 under pooled CER and ch7 under macro CER are posthoc static comparators, not deployable development-selected winners.

AliMeeting RMS improved 145 sources and worsened 65 against ch4, including damage to 21 of 240 baseline-perfect sources. EV damaged 10 of 263 RMS-perfect sources. Gains therefore do not establish harmless input selection. The S

ch0-to-RMS supplement saves only 16 edits, with intervals crossing zero; date 20200702 contributes 19 net saved edits, and omitting it reverses the macro difference to +0.04428 percentage points. This remains a descriptive sensitivity, not another confirmation.

In S, all five edits separating revised and legacy probing arise from same-wave fallback, rather than distinct-input evidence. Of 600 sources, 537 are single-part in both views and 22 have differing split boundaries. A distinct-agree multipart case, `20200623_S_R001S01C01_u00672`, has first/Qwen errors 5/1, so stopping misses four repairs. Matching-boundary agreement can also miss a repair (`20200623_S_R001S03C01_u00517`, 2/1), or prevent teacher harm (`20200630_S_R001S07C01_u00189`, 6/9). Distinct probing and cheap same-view also differ on the identical-input fallback branch, so their comparison does not isolate the information value of a paid view. These corpus-reference examples do not establish audio-adjudicated semantic correctness or safety.

The measured paid-probe increment over Mean95 is modest and quality differences remain unresolved. Cheap identity rules and both confidence-mixture working points constrain any distinctive-benefit claim. An untested two-SV confidence ensemble is not declared inferior. Outcomes on one device, two specific exports, supplied speech boundaries, sparse date strata, and uncertain participant dependence do not establish universal routing optimality, equivalent quality, or cross-hardware superiority.

S12. Author-owned declarations

The existing materials state that the experiments use existing corpora and include no new participant recordings. This factual scope is not an institutional ethics approval or exemption. The following declarations distinguish author-confirmed information from remaining factual items:

- **Author:** OUYANG Tiancheng is the sole author and corresponding author.
  Independent Researcher; nighty@nightytech.com.
  ORCID: 0009-0002-0335-7147.
- **Funding:** This research received no external funding.
- **Competing interests:** The author declares no competing interests.
- **Ethics and data use:** This study used existing AliMeeting and AISHELL-4 corpus recordings and their supplied annotations. No new participants were recruited and no new recordings were collected. Data access and attribution follow the terms published by the original corpus distributors. No institutional approval or exemption is claimed.
- **Model acquisition and conversion provenance:** S3 reports recovered Qwen materials, matched deployment hashes, and unresolved conversion details. The author confirms original SenseVoice Small; its exact historical conversion chain remains unavailable.
- **Availability:** The companion repository and its access boundaries are specified in S10.
- **Publication status and approval:** The sole author approves submission. The manuscript has not been published and is not under consideration elsewhere; no related prior version was reported by the author.

The companion `AUTHOR_CHECKLIST.md` lists the factual declarations and remaining source details to complete before submission.

S13. Complete additional figures

The figures below retain the static-channel and full paired-comparison displays removed from the condensed main article. They reproduce existing results; no new analysis or quality sample is added.

Figure S1(a,b) shows all eight fixed channels followed by RMS on the same 600 exposed AliMeeting sources and 18 recording files, with 13,990 normalized reference characters. They display existing point estimates in channel-index order; paired-difference intervals are not repurposed as single-policy intervals, and no phase timing is merged.

Figure S2 retains all nine distinct-probe-versus-comparator contrasts: recording-macro and pooled CER differences, and candidate/comparator actual-service ratios on a logarithmic axis. Quality uses 600 sources; actual cost uses the pre-selected 120-source subset, both from the same 30 recording files. Intervals are marginal descriptive 95%, conditional on observed date-by-R strata; a singleton stratum contributes no estimated within-stratum resampling variance. Blue markers identify protocol-named contrasts. These are not familywise-protected nine-comparison intervals, quality equivalence/noninferiority, or a joint quality-cost region. The figure's exact numeric companion is `figures/figS2_all_paired_contrasts.source.csv` (relative to this supplement source).

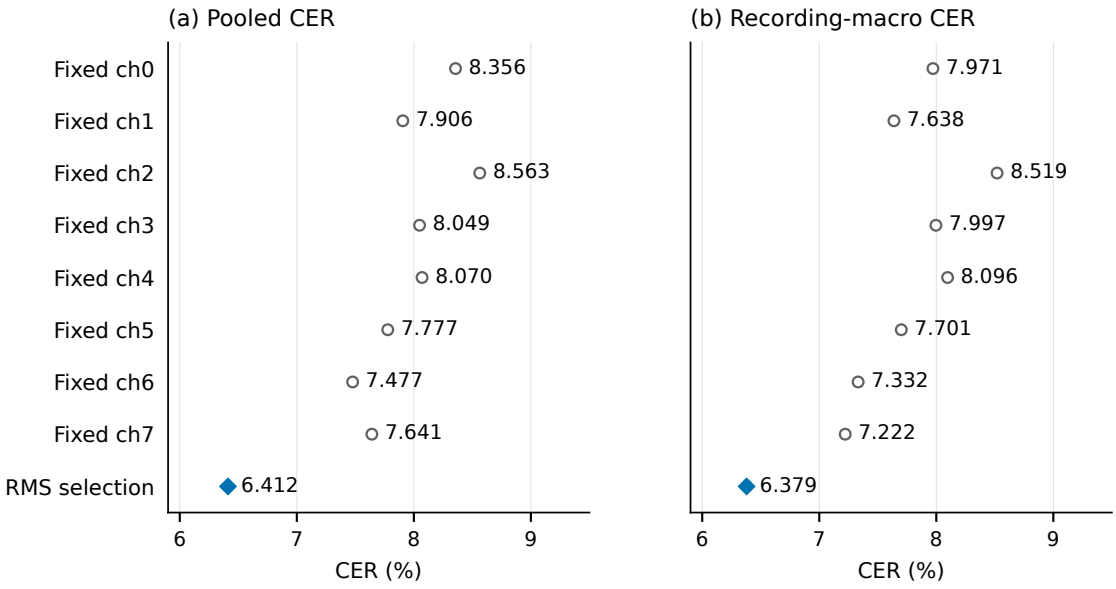


Figure S1: Complete AliMeeting static controls: pooled and recording-macro CER. Same 600-source exposed cohort; no new independent quality samples or merged costs.

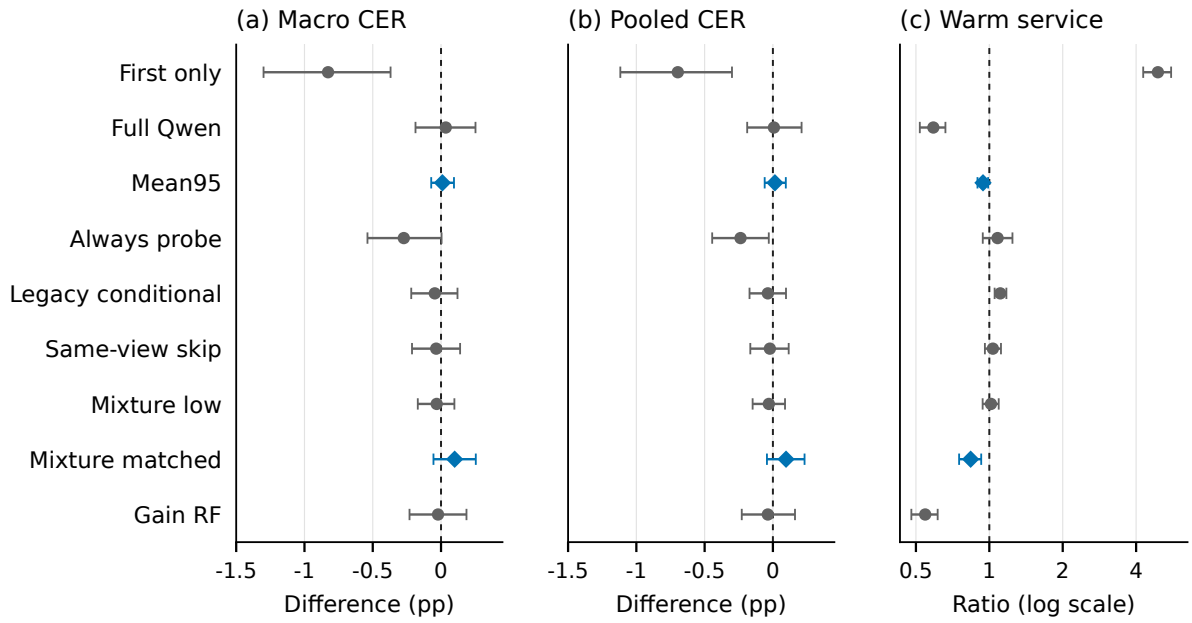


Figure S2: Complete S probe contrasts. CER is Distinct minus the row comparator (pp); service is Distinct/comparator. Intervals are marginal descriptive 95%; quality uses 600 sources and service 120. The ratio axis is logarithmic.

S14. Posthoc audit sensitivities added during manuscript review

These checks use the already exposed S responses and execution records. They add no recognition, independent source, policy tuning, or confirmatory comparison. The historical ten-policy results and their frozen-assignment intervals remain unchanged. Scripts, input hashes, outputs and replay instructions are retained in `analysis/methods_sensitivity/` of the author's archive.

Mixture allocation. For source i, component probabilities w_k, and the frozen component decision d_{ik}, exact integration gives

$$\mathrm{E}[e_i] = \sum_k w_k \left[d_{ik} e_{Q,i} + (1 - d_{ik}) e_{SV,i} \right].$$

An empty first output follows the frozen escalation rule. This expectation is conditional on the observed response bank and averages component allocation only; it is not sampling uncertainty or expected live service.

Table S9. Quality sensitivity to component allocation.

Mixture	Saved-allocation edits	Expected edits	Expected macro CER (%)
Low	1342	1333.498	10.42215
Matched	1325	1328.805	10.37394

The low mixture's point ordering relative to distinct probing (1338 edits; macro CER 10.46120%) reverses when allocation is integrated out. The main table is not erroneous: it reports the saved allocation actually used for timing. This diagnostic shows that a small quality ordering is not allocation-invariant. No expected service, quality–cost pairing for the expectation, or confidence interval is assigned to these expectations.

Timer containment. Inspection of the author's study archive validates 2,400 actual cells and 1,238 Qwen SDK calls. Its extracted timer primitives support reaggregation but do not independently reverify raw extraction, complete SDK payloads, or native execution. Let S be outer policy service, T the helper's recorded microservice, and W the returned engine-loop duration. Let R be S minus all separately timed stages, including T. For Qwen-called cells, the nonnegative envelope $R + T - W$ conservatively bounds helper hash/evidence work outside the engine loop, together with other residual work; it is not an estimate of pure audit time. Set this teacher-specific envelope to zero for cells that do not call Qwen; this does not set their total instrumentation overhead to zero. Average repeats within source, sources within recording, and recordings equally across the complete 120-source panel.

Table S10. Partial timer envelopes, not corrected service.

Policy	Observed service (s)	Outside-loop envelope (ms)
Full-Qwen	3.046872	2.003
Mean95	1.906102	1.017
Distinct probe	1.795795	0.891

Removing only nonnegative time within those partial envelopes would leave a distinct-probe versus Mean95 point saving between 5.737% and 5.834% in the observed traces. This is an arithmetic sensitivity interval, not a statistical confidence interval or an uninstrumented benchmark. Callback-event journaling and other work inside the SDK/engine window are not fully isolated: events have a timestamp, not paired entry/exit times. Broad containment bounds include genuine decoder work and are too loose to establish the saving after removing all instrumentation. Removing code may also change scheduling or cache behavior. Consequently, only the observed instrumented service is claimed; no corrected production time is estimated or substituted.

S15. Additional paired results and engineering interpretation

S15.1. Estimands, resampling, and independence boundaries

These posthoc analyses use completed, already exposed S response-bank and instrumented execution records. They add no recognition run, new quality source, threshold selection, or independent confirmation. Quality uses 600 sources from 30 full dated recording files and 13,498 normalized reference characters. Cost uses the preselected 120-source subset of those same recordings, with two actual executions per policy and source. The two cohorts retain separate denominators.

Let e_{rp} and n_r be the recording's summed edits and reference characters for policy p. Recording-macro CER averages $100e_{rp}/n_r$ equally across recordings; pooled CER is $100\sum_r e_{rp}/\sum_r n_r$. Service first averages repeated executions within source, then sources within recording, then the 30 recordings equally. Negative paired differences indicate fewer errors or less service. Macro/pooled CER differences are percentage points (pp); additive service differences are milliseconds. A relative service change uses a ratio of aggregate means, not the average of individual request ratios.

Frozen within-stratum bootstrap. Sort complete recording IDs, then resample n_h paired recordings with replacement within each date×R stratum containing n_h recordings. All policies, source-level repeats, and components travel together. Each draw retains the observed $n_h/30$ stratum weight. The 20200808 singleton contributes no within-stratum resampling variance. The fixed seed is 2026101622, with 10,000 draws and linear 2.5th/97.5th percentiles. Recalculated original S points and contrast endpoints match their frozen values to numerical tolerance; those frozen artifacts remain unchanged.

Whole-date cluster sensitivity. Independently initialize the same seed and draw nine of the observed dates uniformly with replacement. Carry every recording of each drawn date, including both R labels if present. If date d is drawn m_d times, each of its recordings has weight $w_r = m_{d(r)}$. For each draw calculate

$$\mathrm{CER}^*_{macro} = \frac{\sum_r w_r(100e_{rp}/n_r)}{\sum_r w_r}, \qquad \mathrm{CER}^*_{pooled} = \frac{100\sum_r w_r e_{rp}}{\sum_r w_r n_r}, \qquad \bar{t}^*_p = \frac{\sum_r w_r t_{rp}}{\sum_r w_r}.$$

The denominator is the weighted number of recordings, not nine or a fixed 30. The actual draws contain 17–43 recordings. Observed points remain the original estimates, not averages of bootstrap draws. Fixed-panel edit counts receive no interval for a changing-size resampled panel.

Table S11. Observed date clusters and source counts.

Date	Recordings	Quality sources	Timing sources
20200623	5	100	20
20200624	5	100	20
20200630	4	80	16
20200701	4	80	16
20200702	4	80	16
20200805	2	40	8
20200806	2	40	8
20200807	3	60	12
20200808	1	20	4

Each recording contributes 20 quality sources and four timing sources. All filename-R labels are R001; the 20200808 stratum is a singleton.

All intervals are marginal, descriptive 95% intervals without simultaneous coverage or a noninferiority claim. The fixed-stratum procedure conditions on the observed dates; the whole-date procedure is an exploratory sensitivity with only nine clusters. Neither establishes new-room performance. Filename R labels, recording files, dates, and policy repeats do not verify independent meetings, speakers, or physical rooms. Recording exchangeability and date exchangeability are distinct working assumptions, not experimentally established facts.

S15.2. Four-arm attribution and quality tradeoffs

Use M=Mean95, D=Distinct, C=Cheap, and L=Legacy. The two policy dimensions are paid probing versus no paid probing, and handing the same-waveform fallback branch to Qwen versus retaining SV. Thus (M, D) use teacher fallback and (C, L) retain SV. These are contrasts of frozen policies, not a randomized factorial experiment or isolated acoustic manipulations. Define

$$\Delta_P = \{(D - M) + (L - C)\}/2, \quad \Delta_F = \{(M - C) + (D - L)\}/2, \quad I = (D - M) - (L - C).$$

The interaction is additive on the stated CER or time scale; a ratio-scale interaction is not substituted.

Table S12. Frozen four-arm policy results and full-Qwen control.

Policy	Edits	Macro CER (%)	Pooled CER (%)	Service (s)
Mean95	1336	10.45095	9.89776	1.906102
Distinct	1338	10.46120	9.91258	1.795795
Cheap	1341	10.49664	9.93481	1.736742
Legacy	1343	10.50689	9.94962	1.619269
full-Qwen	1337	10.42658	9.90517	3.046872

Quality: 600 sources and 13,498 reference characters; service: 120 sources with two actual repeats. Separate estimands, not a joint quality–cost confidence region.

Table S13. Quality simple effects, main effects, and additive interaction (pp).

Effect	Weighting	Point	Date×R 95%	Whole-date 95%
Probe, teacher fallback	Macro	+0.01025	[−0.07184, 0.09486]	[−0.12548, 0.12982]
	Pooled	+0.01482	[−0.06021, 0.09457]	[−0.11516, 0.12082]
Probe, SV fallback	Macro	+0.01025	[−0.07184, 0.09486]	[−0.12548, 0.12982]
	Pooled	+0.01482	[−0.06021, 0.09457]	[−0.11516, 0.12082]
Teacher fallback, no probe	Macro	−0.04569	[−0.21807, 0.12065]	[−0.29978, 0.20452]
	Pooled	−0.03704	[−0.17122, 0.09592]	[−0.27896, 0.21941]
Teacher fallback, probe	Macro	−0.04569	[−0.21807, 0.12065]	[−0.29978, 0.20452]
	Pooled	−0.03704	[−0.17122, 0.09592]	[−0.27896, 0.21941]
Probe main effect	Macro	+0.01025	[−0.07184, 0.09486]	[−0.12548, 0.12982]
	Pooled	+0.01482	[−0.06021, 0.09457]	[−0.11516, 0.12082]
Teacher main effect	Macro	−0.04569	[−0.21807, 0.12065]	[−0.29978, 0.20452]
	Pooled	−0.03704	[−0.17122, 0.09592]	[−0.27896, 0.21941]
Additive interaction	Macro	+0.00000	[0.00000, 0.00000]	[0.00000, 0.00000]
	Pooled	+0.00000	[0.00000, 0.00000]	[0.00000, 0.00000]

Intervals are marginal descriptive percentiles. Main effects average the two corresponding simple effects; identical zero quality interactions follow the checked source-level identity.

Across all 600 sources, probing adds 2 edits under either fallback setting, while teacher fallback changes edits by -5 under either probe setting. Both equalities hold source by source. The identically zero quality interaction is therefore structural for these policies and sources; a zero-width interval does not demonstrate general factor independence or safety. Distinct−Cheap changes total edits by -3: fallback repair and probe omissions must both enter the explanation.

Ordinary Mean95 saves 37.441% of observed service versus full-Qwen. Its macro CER difference is +0.024369 pp, whereas its pooled difference is −0.007409 pp: the signs differ because the recording and character weights differ. Delete-one-date Distinct−Mean95 macro differences span −0.026798 to +0.062731 pp; this diagnostic range is not an interval estimate. For mixtures, retain the timed fixed allocation separately from integration over the frozen component probabilities (low: saved 1342, expected 1333.498 edits; matched: saved 1325, expected 1328.805 edits). These expected edits condition on the observed bank and are not accompanied by invented expected live service.

S15.3. Observed service, additive accounting, and trigger sets

Table S14. Service effects on the additive millisecond scale.

Effect	Point (ms)	Date×R 95%	Whole-date 95%
Probe, teacher fallback	−110.307	[−202.574, −21.488]	[−221.264, −4.957]
Probe, SV fallback	−117.473	[−213.180, −25.055]	[−232.531, −8.020]
Teacher fallback, no probe	+169.360	[75.087, 269.800]	[64.001, 288.669]
Teacher fallback, probe	+176.526	[84.063, 274.392]	[65.712, 297.588]
Probe main effect	−113.890	[−207.902, −23.729]	[−227.560, −7.420]
Teacher main effect	+172.943	[79.656, 271.285]	[65.159, 292.957]
Additive interaction	+7.166	[−5.201, 19.295]	[−7.484, 18.566]
Distinct−Cheap	+59.053	[−72.929, 192.670]	[−109.878, 237.037]
Mean95−full-Qwen	−1140.770	[−1348.505, −927.263]	[−1728.371, −730.814]

Table S15. Disjoint observed cost ledger and paired Distinct−Mean95 differences (ms).

Component	Mean95	Distinct	Difference	Date×R 95%
Frontend	5.274	5.565	+0.291	[−0.038, 0.637]
First recognizer	358.063	358.477	+0.414	[−6.662, 7.285]
Necessary confidence	24.613	24.769	+0.157	[−0.339, 0.660]
Probe recognizer	0.000	68.889	+68.889	[44.513, 94.157]
Probe frontend	0.000	0.054	+0.054	[0.035, 0.074]
Teacher	1517.286	1337.244	−180.042	[−285.647, −78.998]
Routing	0.072	0.079	+0.007	[−0.042, 0.057]
Unassigned residual	0.794	0.717	−0.077	[−0.130, −0.028]
Total service	1906.102	1795.795	−110.307	[−202.574, −21.488]

The first eight rows add to total service before rounding. Marginal interval endpoints do not add. SDK/engine timers are nested inside teacher and are not additional ledger rows.

For Distinct−Mean95, reduced teacher time pays for the extra probe and its frontend; the other components plus residual sum to +0.792 ms. The resulting point saving is 5.787%. The residual is unassigned observed wall time, not identified removable instrumentation. Necessary confidence computation remains part of the policy. Nested SDK, decoder-loop, and observer timers cannot be added to the outer teacher interval or to each other.

Table S16. Observed teacher-time contributions by paired trigger set.

Baseline	Trigger set	Sources	Contribution (ms)	Date×R 95%
Mean95	Both upgrade	54	+4.651	[1.095, 8.239]
Mean95	Baseline only	7	−184.693	[−290.958, −83.121]
Mean95	Distinct only	0	+0.000	[0.000, 0.000]
Mean95	Neither	59	+0.000	[0.000, 0.000]
Cheap	Both upgrade	47	+2.942	[−1.222, 7.260]
Cheap	Baseline only	7	−183.453	[−288.926, −81.955]
Cheap	Distinct only	7	+167.210	[73.807, 266.492]
Cheap	Neither	59	+0.000	[0.000, 0.000]

Each source appears in both rounds (twice as many paired cells). Contributions zero-fill nonmembers on the full 120-source roster, then average repeats, sources within recording, and recordings equally. They are not conditional subgroup service means.

The common-upgrade difference reflects the actual executions and their run/order variation; it is not a causal duration saved by removing a call. Against Cheap, Distinct and Cheap each make 54 teacher calls per round but choose different sources. Equal call counts therefore do not determine equal service. The Distinct−Cheap net service difference also includes the paid probe, and does not establish a unique superior operating point. No counterfactual is estimated by multiplying a call count by a global average duration. Interaction intervals cross zero, but do not establish that interaction is practically negligible.

S15.4. Workload composition, repeated timing, and first requests

Table S17. Composition of the quality bank, timing subset, and complement.

Characteristic	Quality 600	Timing 120	Other 480
Sources	600	120	480
Recordings	30	30	30
Mean duration (s)	5.569	5.284	5.640
Maximum duration (s)	14.965	11.291	14.965
RMS multipart sources	63	7	56
Matching split sources	578	118	460
Same-waveform aliases	202	38	164
Empty first output	2	0	2
Nonempty confidence sources	598	120	478
Mean nonempty confidence	0.916362	0.916511	0.916325
Mean95 teacher fraction (%)	55.83	50.83	57.08
Distinct teacher fraction (%)	48.17	45.00	48.96
Cheap teacher fraction (%)	47.00	45.00	47.50

Routing fractions are response-bank projections on each roster, not measured 600-source calls or inferred timing. Duration and composition summaries are source-weighted; primary CER and service retain their stated recording weights.

The timing subset has shorter speech intervals and fewer multipart inputs. Similar mean confidence does not make the 120-source workload representative of the complete quality bank. Its response-bank routing composition is reported separately from actual execution counts, and no 600-source service estimate is obtained by scaling up the measured subset.

Table S18. Service means by actual timing round and actual calls per round.

Policy	Round 1 (s)	Round 2 (s)	Qwen calls	SV attempts
Mean95	1.872953	1.939251	61	127
Distinct	1.778092	1.813498	54	153
Cheap	1.711886	1.761598	54	127
Legacy	1.592127	1.646411	47	153
full-Qwen	2.977527	3.116217	120	0

Both call-count columns are per round and happen to agree across these two rounds. Multipart decoding makes SV native attempts differ from the 120 policy requests.

Table S19. Paired sensitivity to omission of each round's first scheduled source.

Round	Comparison	Sources	Difference (ms)	Date×R 95%
1	D−M	120	−94.861	[−184.273, −8.110]
1	D−M	119	−95.849	[−185.444, −9.388]
1	D−C	120	+66.206	[−61.552, 196.988]
1	D−C	119	+65.242	[−62.807, 195.974]
2	D−M	120	−125.752	[−222.701, −33.235]
2	D−M	119	−135.685	[−236.873, −38.171]
2	D−C	120	+51.901	[−85.581, 188.975]
2	D−C	119	+41.780	[−99.372, 183.782]

For each comparison the 120-source result precedes the 119-source sensitivity. Omit the same source from both arms; reaverage its recording's remaining three sources, retaining all 30 equally weighted recordings. This is not an estimated cold-start correction.

The first formal requests were round 1: `20200701_S_R001S05C01_u00163` (full-Qwen, 3.716179 s); round 2: `20200624_S_R001S05C01_u00682` (gain-RF, 3.444386 s). These are different sources and policies; comparing their durations does not identify a cold-start effect. Distinct's observed savings versus Mean95 in the complete rounds are 5.065% and 6.485%. Delete-one-date service differences span −133.370982 to −54.031726 ms.

On the same 120 sources, both actual rounds match their corresponding bank projection on text, actions, teacher decisions, and edit counts. Mean95 and Distinct each have 253 edits out of 2,551 reference characters, but the macro CER difference is −0.029295 pp. This subset diagnostic neither replaces the 600-source quality contrast nor establishes quality equivalence. The two rounds reuse the same source set; they are not additional independent recordings.

S15.5. Measured runtime boundary and reproducibility

The historical service clock starts with the multichannel input buffer ready and ends after policy output and the recorded completion boundary. Model initialization, gate waits, and post-service archiving are outside that clock. Both recognizers are initialized once in each timing process. The bound initializer has no explicit recognition warm-up loop; dedicated inference warm-up sources, order, and count are not recorded. Thus "warm" here means initialized and loaded, not demonstrated steady-state latency or verified physical memory residency. Constructor-internal warm-up is not established.

Qwen uses the fixed 30-second encoder configuration; each SDK invocation creates a fresh stream session and clears mode-0 context, with prefix cache disabled. The supplied speech intervals are much shorter than 30 seconds, so the reported pipeline comparison depends on these exports and their fixed encoder work. The configuration field `enabled_cpus=4` is not an affinity mask. Initialization snapshots and each round's first final gate-ready snapshot show CPU 0–7; the other 1,199 gate-ready snapshots and all 1,200 post-archive snapshots per round show `[4]`. This is the sampled calling thread's affinity; it does not verify all SDK workers or continuous CPU use throughout service.

Table S20. Historical final gate-ready snapshots; 1,200 cells per round.

Measurement	Round 1	Round 2
Gate-ready load1, min–max	1.01–3.99	1.06–3.99
Gate-ready maximum sensor, °C	69.307–74.846	71.153–74.846
Gate-ready VmSwap (kB), min–max	115784–471924	646336–1507820
Nonzero-swap gate snapshots	1200/1200	1200/1200
Total gate wait outside service (s)	870.038	895.040
Cells with positive gate wait	173	179

VmSwap is allocated process swap, not page-fault activity or an identified per-request delay. Snapshot ranges do not certify conditions throughout service; wait times are excluded from the service endpoint.

All 2,400 final gate-ready snapshots meet the recorded 75°C/load1≤ 4 conditions. Post-initialization and post-archive snapshots need not meet them. The latter include both service and archive work, so they cannot attribute temperature/load changes to service alone. The temperature is the maximum reported sensor value in each snapshot, not continuous in-service telemetry. Allocated swap is nonzero; neither swap-free execution nor a swap-induced latency penalty is demonstrated. No power or energy result follows from these timing records.

These exclusions prevent interpreting service as capture-to-text latency, ungated throughput, or a deployment SLA. The tables reconstruct completed instrumented executions. They do not measure a minimal-instrumentation mode or estimate its latency by subtracting the residual. The partial timer envelopes in S14 retain their original, narrower interpretation.

Rebuild and provenance. From the delivered revision root, run

```
python3 -B analysis/jaes_statistics_v1/build_revision_tables.py --validate-replay
```

The builder verifies completed upstream output hashes, generates this single S15 fragment and its table-row index, and records every consumed input hash. The companion `--check` mode checks current generated bytes without writing. The binding and actual build receipt are in `analysis/jaes_statistics_v1/tables/`. A relocated rebuild checks the fragment and three deterministic companion files byte for byte. Original statistics and cost-analysis receipts retain their own executed-analysis identities; table generation does not rerun bootstrap, recognition, or original A/C/F analyses.

S15.6. Fixed conditional mechanism and limits of interpretation

This inherited diagnostic retains the 127 nonempty, non-alias sources in the fixed $[.90, .95)$ confidence range and its existing 0.925 split. No bin boundary or policy parameter was selected by this table build. "Agreement" compares the first and second SV outputs; teacher gain compares Qwen with first-pass RMS-SV. Their different roles must not be conflated.

Table S21. Inherited fixed conditional diagnostic, retaining all eight cells.

Split	Band	Agree	Sources	Ref. chars	Help/harm/tie	Saved edits	Gain (pp)
S	Low	Yes	11	176	2/1/8	$+1$	$+0.568$
S	Low	No	31	628	14/8/9	$+6$	$+0.955$
S	High	Yes	34	748	5/7/22	-3	-0.401
S	High	No	34	851	13/10/11	$+5$	$+0.588$
O	Low	Yes	1	33	1/0/0	$+4$	$+12.121$
O	Low	No	9	416	6/3/0	$+9$	$+2.163$
O	High	Yes	0	0	0/0/0	$+0$	—
O	High	No	7	318	4/2/1	$+4$	$+1.258$

S: both inputs have one part with matching boundaries; O: other split conditions. Low $= [.90, .925)$, High $= [.925, .95)$. Gain $= 100(e_{SV} - e_Q)/n$; positive means teacher repair. A zero-denominator cell has undefined gain (—), not zero effect. The one-source and empty cells remain visible.

Table S22. Disagreement minus agreement teacher gain; delete-date points (pp).

Split	Band	Observed	Delete-date range	Defined deletions
S	Low	$+0.387$	-0.398 to $+1.389$	8
S	High	$+0.989$	$+0.285$ to $+2.367$	8
O	Low	-9.958	-11.526 to -7.907	7
O	High	—	—	0

The diagnostic spans eight related dates, not nine independent replications. Ranges summarize defined point estimates only; they are not confidence intervals or adjusted causal effects.

Among sources with both inputs single-part and matching splits, disagreement has a larger observed normalized teacher gain than agreement in both confidence bands. However, the low-band direction reverses when 20200808 is deleted. The other split conditions have just one low-band agreement source and no high-band agreement source. Retaining the sparse and undefined cells makes this support boundary visible; no zero is imputed for a missing comparison.

The conditional table holds a coarse confidence band and one segmentation condition fixed, but does not balance continuous confidence, content, speakers, acoustic quality, or selection into the diagnostic. Altering a waveform can also alter its segmentation. These associations therefore do not isolate pure spatial information, establish acoustic causality, prove that agreement is safe, or validate a new routing threshold. Eight diagnostic-related dates are a subset of the nine-date S cohort, not independent replications of a finding.

The complete bank contains 181/123/296 teacher help/harm/tie sources, respectively, with 95 net edits saved by Qwen relative to first-only SV. The net result includes harmful upgrades as well as useful repairs; potential repairs skipped by probing and harmful upgrades avoided by probing both remain in the all-source case index. Illustrations and full source IDs are retained in `analysis/jaes_cost_v1/CASE_INDEX.csv` and its companion `analysis/jaes_cost_v1/representative_cases.csv`; selected examples are not a representative sample or a new semantic gold standard.

For an engineer using these exports, the observed choices separate three questions: whether ordinary cascading buys enough service reduction relative to all-Qwen, whether same-waveform fallback repairs justify their measured cost, and whether a paid second view improves the chosen quality–service tradeoff relative to both Mean95 and Cheap. The four-arm decomposition answers those questions for the measured workload without assigning fallback repairs to spatial information. Applying the ranking to a new workload requires its own measurements; the present posthoc conditionals provide hypotheses and failure cases rather than a validated predictor of which meeting will benefit.

S16. Complete input tables and plots

Table S23: All-fixed supplement, same 600 AliMeeting sources.

SV input	Pooled CER (%)	Macro CER (%)
ch0	8.35597	7.97140
ch1	7.90565	7.63802
ch2	8.56326	8.51925
ch3	8.04861	7.99690
ch4	8.07005	8.09612
ch5	7.77698	7.70087
ch6	7.47677	7.33205
ch7	7.64117	7.22176
RMS	6.41172	6.37927

Table S24: C calibration quality; C1 prefixes excluded.

Policy	Edits	Pooled CER (%)	Macro CER (%)
ch0	2648	12.9842	13.1675
E1 startup energy	2638	12.9352	13.1422
L1 labelled startup	2621	12.8518	13.0393
Dynamic RMS	2597	12.7341	12.9281
Dynamic EV	2608	12.7881	12.9874

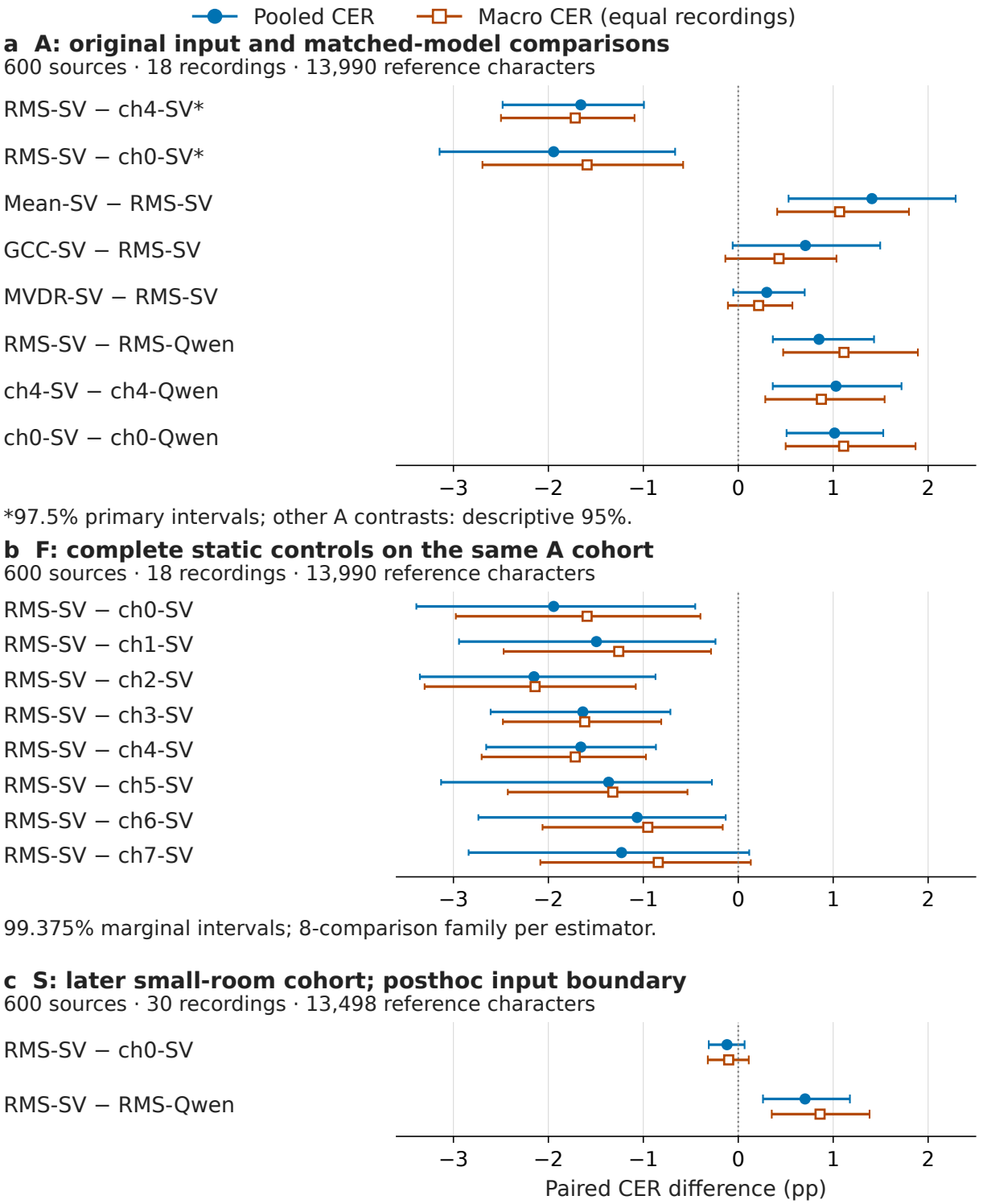


Figure S3: Complete paired input and recognizer effects, retaining cohort and interval-family identities. Refer to the plotting data and manifest for every contrast.

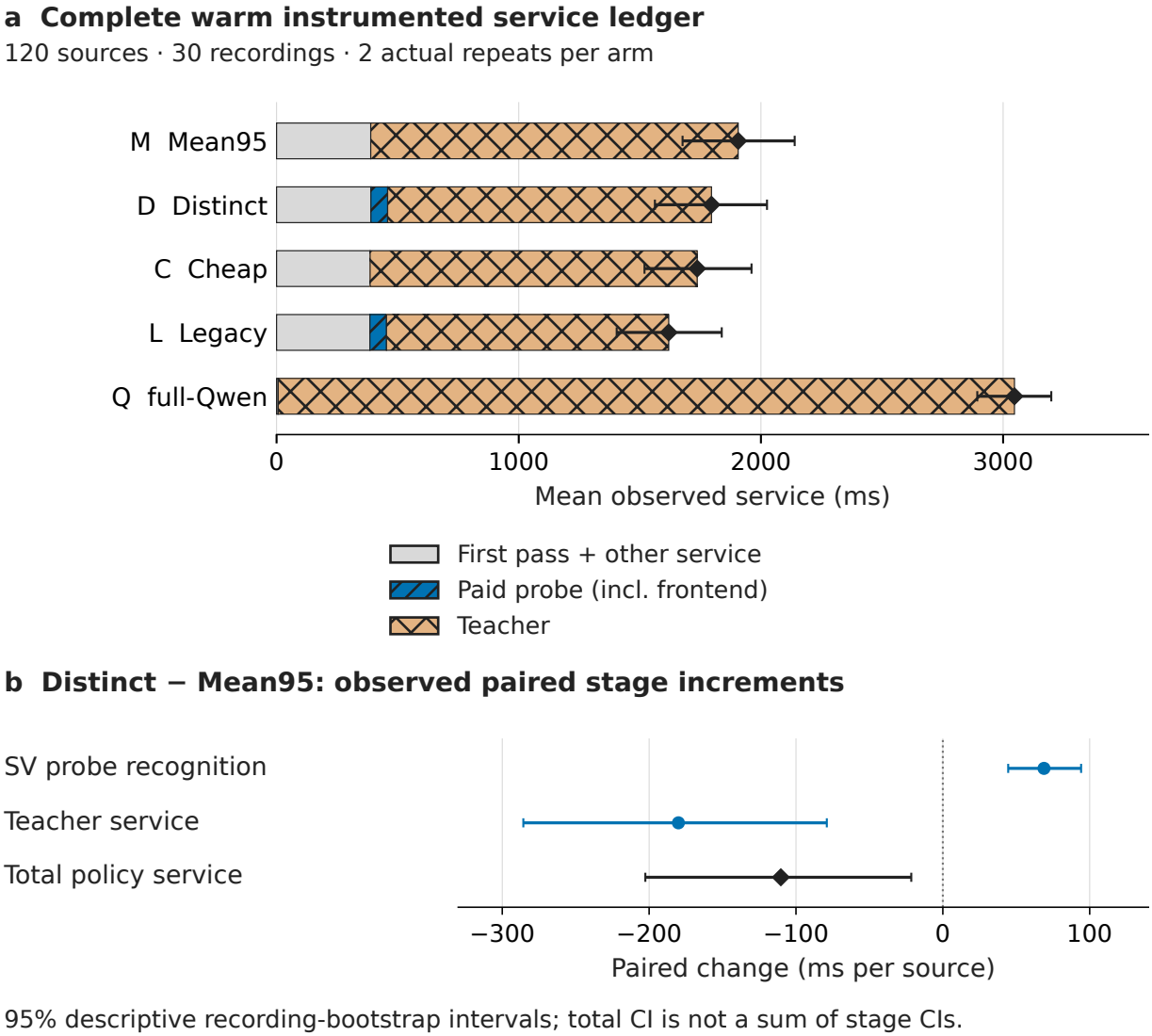


Figure S4: Observed instrumented component ledger and paired Distinct-minus-Mean95 costs. Nested SDK timers are not added to outer teacher service; uncertainty describes paired recording resampling.

S17. Revision provenance

The revision dated November 11, 2026 clarifies the interpretation and availability of the existing results; it adds no experiments. The original frozen comparisons and their exposure identities remain unchanged. Analysis and plotting code, input/output hashes, software versions, and source/PDF build records are retained in the author's archive. The five scientific figure sets are generated from numeric CSV/JSON inputs with Matplotlib. The JAES2026.08 class is used with licensed fonts; the bibliography style has one local change to spell out "in Proceedings of the." Public numerical reproduction is described in S9.

A local callback fixture found an error-acceptance gap in the historical decoder: an ERROR callback followed by FINISH and a zero return code can leave a partial output classified as successful. The matched measurement code retains the error flag in both modes and rejects this sequence; local callback fixtures cover error/finish order and reset behavior. A separate readback of the retained S bank and timing callbacks checked 1,838 actual SDK calls (600 bank, 619 in each timing round). Their saved events contained 29,565 NORMAL states and 1,838 FINISH states, with no ERROR state or flagged return/abort/finish/SDK-failure record. Original full-call hashes match native completion manifests. Replay and extraction receipts are retained in the author's `analysis/callback_audit_v1/` archive. This readback does not establish coverage of all possible SDK errors, unrecorded lower-level failures, or the A/F cohorts. The fixture bug and the absence of that sequence in these S records are separate findings.

The completed audit/minimal follow-up comprises 2,400 actual requests and 1,200 matched products; its four mode blocks and product checks are described in S18. Minimal processes retain zero audit-array bytes. Per-cell gates and observations remain outside service. Time-block, operating-system page-cache and thermal differences remain possible; cross-mode differences do not isolate instrumentation overhead. Raw records and accepted analysis are retained in the author's `raw_execution/matched_v5/` and `analysis/matched_measurement_v2/actual_v5_01/` archives. The follow-up reuses the exposed 120-source timing panel and does not create a new 600-source quality confirmation.

The recovered model materials and remaining conversion gaps are summarized in S3. The public evidence location is provided in S10. The author supplied the biography; any applicable institutional determination and material-specific permissions remain separate author and licensing matters.

S18. Completed matched measurement follow-up

The source roster is unchanged: 120 sources, four per recording, 30 recording files and nine date groups. Four fresh processes execute round-1 audit, round-1 minimal, round-2 minimal, and round-2 audit. Each process uses its own fixed development warm-up; no minimal process runs audit inference. All four completed measurement runs were included in the analysis.

Checks against the author's raw records found that each of the 1,200 same-round/source/policy audit–minimal pairs matches text, confidence, route, selected input, native cuts/tokens, SDK completion/cache state, and generated work. All 2,400 rows match available historical bank text, routing, and native/SDK-count fields, with zero edit changes on the 120-source panel. Historical full-Qwen projections lack waveform-identity fields for 480 rows; missing identity comparisons are not counted as agreement. Complete receipts and raw rows are retained in the author's archive under `raw_executio n/matched_v5/`; original tensor payloads remain indexed separately on the research device. The public numerical extract cannot independently re-establish these product matches.

The field `retained_audit_evidence_before_archive` records logical audit retention after service and immediately before archiving. Its inherited scope string, "service entry," is a labeling error; it is not a service-entry or physical-RSS measurement. Raw fields and their hashes are preserved unchanged. This correction does not change service times, product checks, or estimates.

The primary minimal D–M difference is −100.529 ms [−183.263, −20.045]; whole-date sensitivity is [−201.782, −8.464]. Minimal D–C is +49.384 ms [−69.472, +173.186]. M–Q relative service change is −35.388% [−42.025, −28.666]. The mode-associated D–M difference-in-differences is −24.117 ms [−45.280, −2.780], with whole-date sensitivity [−36.419, −10.356]. These compare distinct time blocks and cache histories; they are not isolated instrumentation costs. No data-derived quality margin or equivalence claim is introduced.

The public command in S9 recomputes the matched means, intervals, component ledger, round and leave-date sensitivities from released numerical rows, comparing all six aggregate CSVs. It calls the frozen analyzer's numerical functions; the original raw-product CLI and figure/table generation scripts require the separate author's archive and are not the public reproduction entry point.

Table S25: Fixed date × R recording bootstrap: arm service means and 95% CIs.

Mode	Policy	Mean (s)	95% CI (s)
Minimal	Q	2.822	[2.694, 2.948]
Minimal	M	1.823	[1.618, 2.032]
Minimal	D	1.723	[1.518, 1.926]
Minimal	C	1.673	[1.473, 1.878]
Minimal	L	1.575	[1.384, 1.766]
Audit	Q	2.834	[2.703, 2.964]
Audit	M	1.831	[1.623, 2.044]
Audit	D	1.755	[1.546, 1.961]
Audit	C	1.697	[1.499, 1.900]
Audit	L	1.610	[1.409, 1.812]

Table S26: Minimal: Fixed date × R recording bootstrap. Candidate minus comparator; relative change is $100(\mu_a/\mu_b - 1)$. Additive interaction: $(D - M) - (L - C)$.

Contrast	Δ (ms)	95% CI (ms)	Change (%)	95% CI (%)
$D - M$	−100.529	[−183.263, −20.045]	−5.514	[−9.938, −1.123]
$D - C$	49.384	[−69.472, 173.186]	2.951	[−4.009, 10.888]
$M - Q$	−998.605	[−1186.289, −806.069]	−35.388	[−42.025, −28.666]
$L - C$	−98.059	[−183.290, −16.518]	−5.860	[−10.661, −1.043]
$M - C$	149.913	[62.859, 241.568]	8.959	[3.686, 15.132]
$D - L$	147.443	[66.091, 233.979]	9.360	[4.198, 15.335]
Probe × fallback	−2.470	[−26.109, 22.246]	—	—

Historical quality: 600 sources (separate bank). Actual service: 120 sources, 30 recordings, two rounds per mode. Repeats are averaged within source, sources within recording, then recordings equally. Intervals are marginal descriptive 95% paired bootstrap intervals (10,000 draws). Q=full-Qwen; M=Mean95; D=Distinct; C=Cheap; L=Legacy. Negative differences mean less service. Mode comparisons retain time-block/cache confounding; no equivalence or pure-overhead conclusion is implied. Interaction percentages are undefined, not zero.

Table S27: Audit: Fixed date × R recording bootstrap. Candidate minus comparator; relative change is $100(\mu_a/\mu_b - 1)$. Additive interaction: $(D - M) - (L - C)$.

Contrast	Δ (ms)	95% CI (ms)	Change (%)	95% CI (%)
$D - M$	−76.412	[−156.570, 0.496]	−4.173	[−8.374, 0.027]
$D - C$	57.706	[−69.677, 188.031]	3.400	[−3.915, 11.699]
$M - Q$	−1002.814	[−1190.224, −808.456]	−35.386	[−42.062, −28.736]
$L - C$	−86.959	[−170.108, −5.958]	−5.124	[−9.979, −0.363]
$M - C$	134.118	[44.277, 228.674]	7.903	[2.563, 13.978]
$D - L$	144.665	[49.310, 244.211]	8.985	[2.990, 15.825]
Probe × fallback	10.548	[−14.107, 35.401]	—	—

Table S28: Fixed date × R recording bootstrap: paired D–M outer ledger. Components add only as point estimates; CI endpoints are not summed. Residual closes service; nested SDK timers are not added.

Component	Minimal (ms)	95% CI	Audit (ms)	95% CI
Front end	-0.008	$[-0.316, 0.315]$	0.059	$[-0.195, 0.308]$
First SV	-1.781	$[-10.984, 7.542]$	11.134	$[-2.279, 24.385]$
First confidence	-0.477	$[-1.351, 0.404]$	0.256	$[-0.904, 1.480]$
Probe SV	72.632	$[47.221, 98.628]$	77.822	$[51.933, 103.882]$
Probe front end	0.065	$[0.041, 0.089]$	0.074	$[0.050, 0.099]$
Teacher (outer)	-171.028	$[-268.199, -77.428]$	-165.685	$[-261.488, -74.043]$
Routing	0.009	$[0.006, 0.012]$	0.010	$[0.006, 0.015]$
Service residual	0.060	$[0.003, 0.170]$	-0.082	$[-0.150, -0.018]$
Total service	-100.529	$[-183.263, -20.045]$	-76.412	$[-156.570, 0.496]$

Table S29: Whole-date sensitivity: arm service means and 95% CIs.

Mode	Policy	Mean (s)	95% CI (s)
Minimal	Q	2.822	$[2.640, 3.056]$
Minimal	M	1.823	$[1.478, 2.053]$
Minimal	D	1.723	$[1.428, 1.937]$
Minimal	C	1.673	$[1.298, 1.906]$
Minimal	L	1.575	$[1.246, 1.777]$
Audit	Q	2.834	$[2.654, 3.068]$
Audit	M	1.831	$[1.480, 2.062]$
Audit	D	1.755	$[1.461, 1.965]$
Audit	C	1.697	$[1.315, 1.935]$
Audit	L	1.610	$[1.287, 1.812]$

Table S30: Minimal: Whole-date sensitivity. Candidate minus comparator; relative change is $100(\mu_a/\mu_b - 1)$. Additive interaction: $(D - M) - (L - C)$.

Contrast	Δ (ms)	95% CI (ms)	Change (%)	95% CI (%)
$D - M$	-100.529	$[-201.782, -8.464]$	-5.514	$[-10.661, -0.507]$
$D - C$	49.384	$[-100.131, 208.233]$	2.951	$[-5.614, 14.953]$
$M - Q$	-998.605	$[-1506.244, -639.164]$	-35.388	$[-50.274, -23.911]$
$L - C$	-98.059	$[-189.140, -4.106]$	-5.860	$[-10.586, -0.264]$
$M - C$	149.913	$[44.275, 266.187]$	8.959	$[2.672, 18.905]$
$D - L$	147.443	$[41.421, 260.938]$	9.360	$[2.601, 19.055]$
Probe × fallback	-2.470	$[-24.011, 18.586]$	—	—

Table S31: Audit: Whole-date sensitivity. Candidate minus comparator; relative change is $100(\mu_a/\mu_b - 1)$. Additive interaction: $(D - M) - (L - C)$.

Contrast	Δ (ms)	95% CI (ms)	Change (%)	95% CI (%)
$D - M$	-76.412	$[-172.973, 12.342]$	-4.173	$[-9.011, 0.788]$
$D - C$	57.706	$[-90.202, 213.457]$	3.400	$[-5.050, 15.205]$
$M - Q$	-1002.814	$[-1516.182, -640.952]$	-35.386	$[-50.360, -23.803]$
$L - C$	-86.959	$[-189.947, 6.602]$	-5.124	$[-10.466, 0.479]$
$M - C$	134.118	$[29.545, 249.188]$	7.903	$[1.723, 17.283]$
$D - L$	144.665	$[48.298, 245.318]$	8.985	$[2.947, 17.482]$
Probe × fallback	10.548	$[-10.437, 28.176]$	—	—

Table S32: Whole-date sensitivity: paired D–M outer ledger. Components add only as point estimates; CI endpoints are not summed. Residual closes service; nested SDK timers are not added.

Component	Minimal (ms)	95% CI	Audit (ms)	95% CI
Front end	-0.008	$[-0.280, 0.233]$	0.059	$[-0.226, 0.295]$
First SV	-1.781	$[-17.648, 10.058]$	11.134	$[0.856, 21.185]$
First confidence	-0.477	$[-1.603, 0.387]$	0.256	$[-1.078, 1.167]$
Probe SV	72.632	$[37.645, 105.263]$	77.822	$[38.907, 113.938]$
Probe front end	0.065	$[0.034, 0.093]$	0.074	$[0.035, 0.107]$
Teacher (outer)	-171.028	$[-283.998, -43.928]$	-165.685	$[-277.340, -39.088]$
Routing	0.009	$[0.003, 0.014]$	0.010	$[0.005, 0.015]$
Service residual	0.060	$[0.003, 0.174]$	-0.082	$[-0.128, -0.035]$
Total service	-100.529	$[-201.782, -8.464]$	-76.412	$[-172.973, 12.342]$

Table S33: Round-specific core contrasts (point estimates only; rounds are not independent recordings).

Mode	Round	Contrast	Δ (ms)	Change (%)
Minimal	1	$D - M$	-92.880	-5.063
Minimal	1	$D - C$	60.159	3.578
Minimal	1	$M - Q$	-994.066	-35.145
Minimal	2	$D - M$	-108.179	-5.970
Minimal	2	$D - C$	38.608	2.318
Minimal	2	$M - Q$	-1003.143	-35.633
Audit	1	$D - M$	-69.172	-3.758
Audit	1	$D - C$	71.665	4.216
Audit	1	$M - Q$	-984.202	-34.842
Audit	2	$D - M$	-83.652	-4.592
Audit	2	$D - C$	43.746	2.582
Audit	2	$M - Q$	-1021.427	-35.926

Table S34: Leave-one-date-out D–M sensitivity (point estimates; no interval supplied).

Mode	Omitted date	Δ (ms)
Minimal	20200623	-111.516
Minimal	20200624	-47.678
Minimal	20200630	-96.913
Minimal	20200701	-121.462
Minimal	20200702	-93.610
Minimal	20200805	-112.121
Minimal	20200806	-107.762
Minimal	20200807	-105.494
Minimal	20200808	-104.500
Audit	20200623	-82.430
Audit	20200624	-26.345
Audit	20200630	-75.390
Audit	20200701	-95.501
Audit	20200702	-69.167
Audit	20200805	-89.969
Audit	20200806	-82.995
Audit	20200807	-84.025
Audit	20200808	-78.147

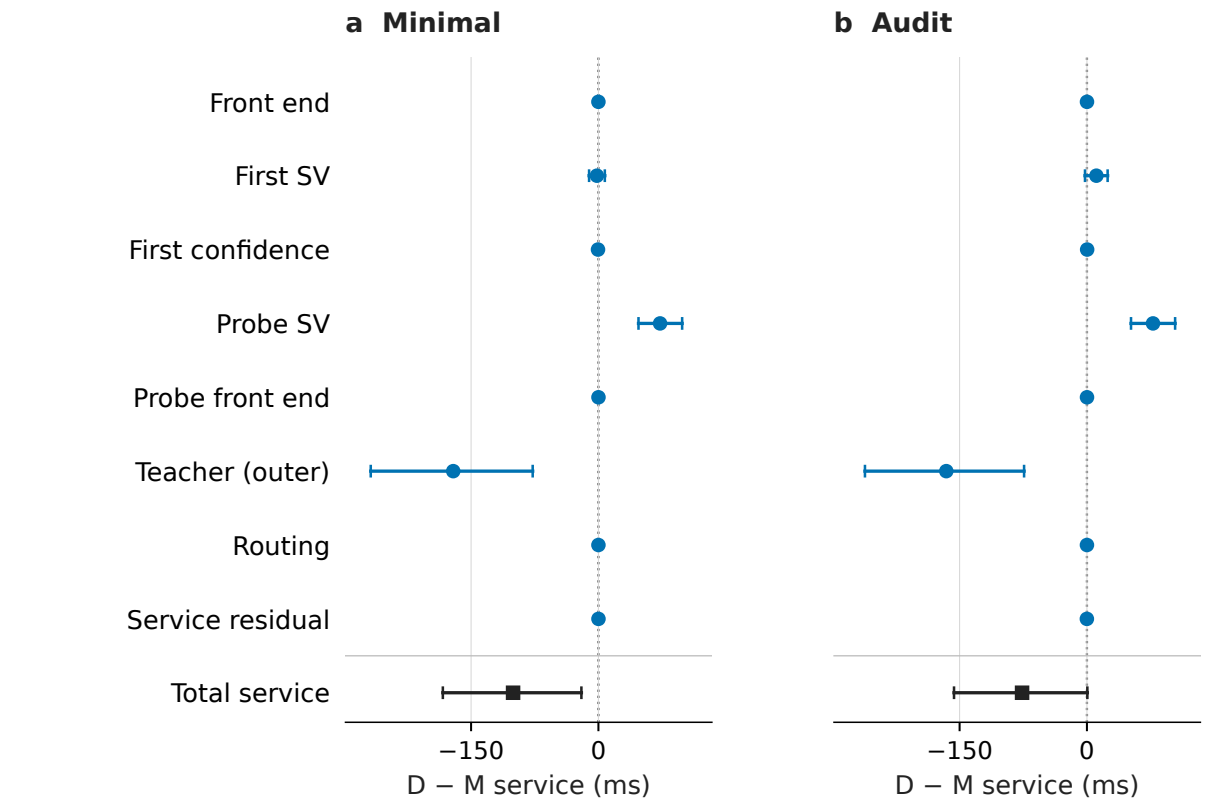


Figure S5: Completed matched follow-up: paired D–M disjoint component costs, ms. Point differences sum to total service; component interval limits are not additive. The residual is observed, not assumed removable overhead. Teacher denotes the enclosing Qwen call timer.

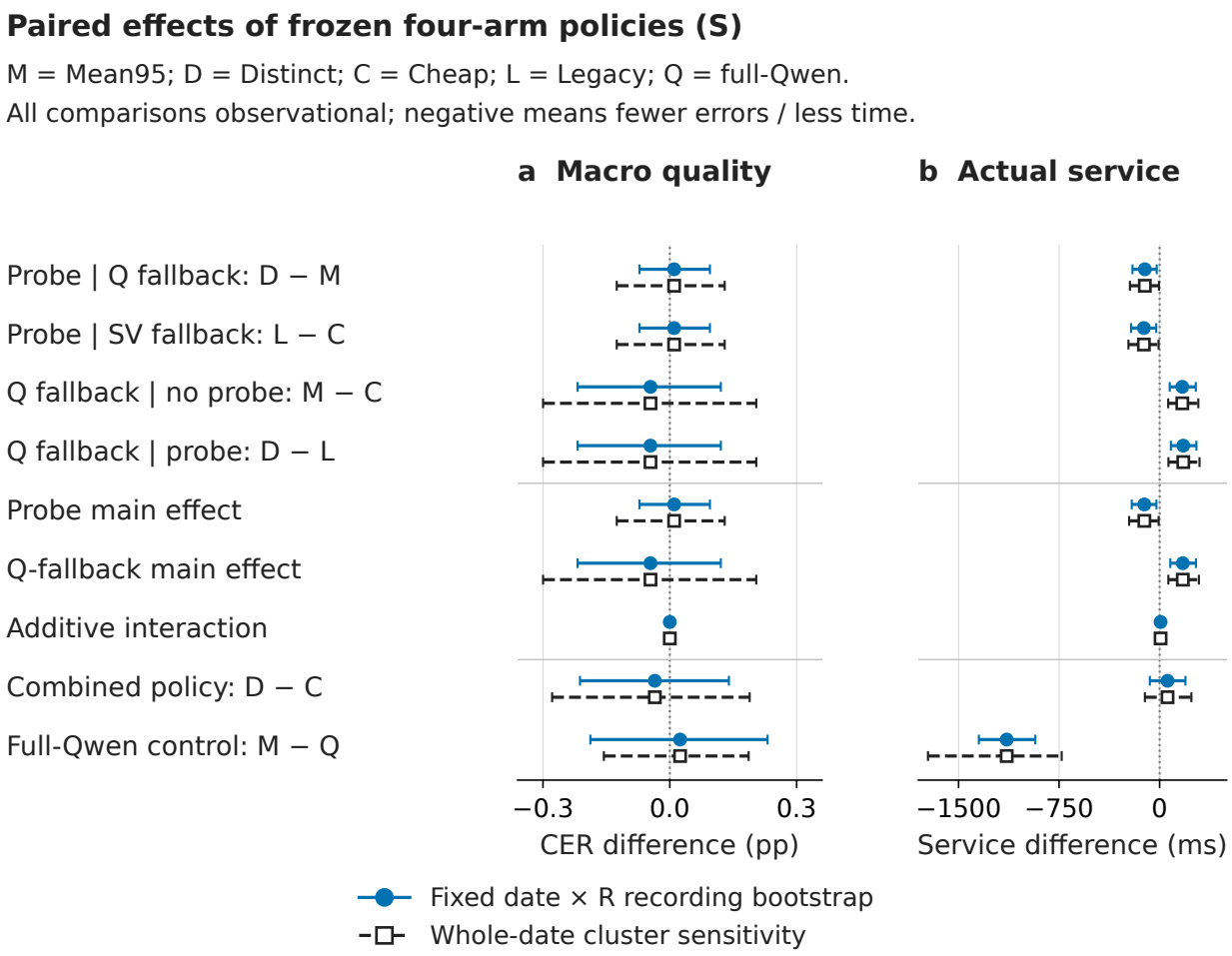


Figure S6: Historical four-policy decomposition with instrumented service. M, D, C, and L denote Mean95, Distinct, Cheap, and Legacy; Q denotes full-Qwen. CER uses 600 bank sources, service 120 sources with two real repeats. Solid intervals are descriptive 95% fixed-date recording bootstraps; dashed intervals are whole-date cluster sensitivity analyses (10,000 draws each). Main effects average the two simple effects; interaction is D−M−L+C. The zero quality interaction is a source-level structural identity.